

DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

B. Country*: the People's Republic of China

Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: The Institute of Scientific and Technical Information of China (abbreviated as ISTIC)
(affiliated to the Ministry of Science and Technology of China)

Website: www.istic.ac.cn

Email address*:

Phone number*:

Please describe the role/mandate of your organization: The Institute of Scientific and Technical Information of China (abbreviated as ISTIC), established on October 2nd, 1956, is a national-level scientific and technological information research institute directly under the Ministry of Science and Technology. It mainly provides comprehensive information services for enterprises, universities, research institutes and scientific research personnel, and provides decision-making support for administrative departments. With application-oriented product development and platform services, it integrates resources, talents and technologies, to strive to build an International Think Tank of Innovation Strategy, the National Science and Technology Information Big Data Center and the National Engineering Center of Science and Technology Information.

Officially designated contact person(s) who completed this survey:

Full name*: ZHANG Yiqing

Position*: Liaison Officer for Multilateral Affairs, Ministry of Science and Technology of China

Email address*:

Full Name(s) of designated official(s) certifying the report: Department of International Cooperation, Ministry of Science and Technology of China

Brief description of the consultation process established for the preparation of the report:

To conduct a comprehensive assessment of the implementation of the 2021 Recommendation, ISTIC, under the commission of its supervising authority, carried out multiple rounds of online surveys and expert consultations. The process involved extensive discussions and deliberations with experts from various research institutions, universities, and relevant social organizations.

Experts analyzed the status of implementation across different sectors in China, providing recommendations for the drafting of the questionnaire. Additionally, a dedicated ISTIC research team conducted online investigations into the key survey questions. After thorough review and consideration, the final report was compiled.

Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*:

Institute of Scientific and Technical Information of China (ISTIC), National Science Library, Chinese Academy of Sciences, UN Consultative Committee on Open Science and Global Partnership of the China Association for Science and Technology (CCOS), Alliance of National and International Science Organizations for the Belt and Road Regions (ANSO), and so on.

Website*: www.istic.ac.cn; www.las.ac.cn; www.cast.org.cn; www.anso.org.cn.

Email address*

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The Ministry of Science and Technology, the Ministry of Education, the China Association for Science and Technology, the Chinese Academy of Sciences, the National Natural Science Foundation of China, as well as NGOS including the Open Science Promotion Consortium, the Alliance of National and International Science Organizations for the Belt and Road Regions (ANSO), the UN Consultative Committee on Open Science, Global Partnership of the China Association for Science and Technology (CCOS), etc.

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES

If yes, which language(s)?

Chinese.

- 1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES.

If yes, please provide details on the activities organized or foreseen and links as relevant.

Recent years, various initiatives and organizational activities have been undertaken to enhance awareness and promote the practice of open science in China:

- 1) **November 2022:** At the 5th World Laureates Forum, Academician Yang Wei, Chair of the UN Consultative Committee on Open Science and Global Partnership of the China Association for Science and Technology (CAST), issued the "Recommendations for Conduct in International Research Collaborations".
- 2) The Hangzhou Declaration was released in the Sixteenth General Conference of The World Academy of Sciences for the advancement of science in developing countries (TWAS).

- 3) **October 23, 2024:** The World Science and Technology Development Forum hosted Thematic Session 3—"Open Science Infrastructures: Building a Collaborative Platform for the Science Decade," during which an open science initiative and vision were unveiled.
- 4) **November 4–5, 2024:** The **758th Session of Xiangshan Science Conference** was jointly organized by the UN Consultative Committee on Open Science and Global Partnership of CAST and the National Science Library of the Chinese Academy of Sciences (CAS), resulting in the "Xiangshan Initiative on Open Science" .
- 5) **November 18–20, 2024:** The National Science Library of CAS hosted the China Open Access Week 2024 , themed "Open Science in Action: Building an Innovative and Inclusive Future Together." This event provided a platform for in-depth discussions on the latest developments, challenges, and practical approaches in open science, attracting over 200 experts, young scholars, and representatives from various sectors.
- 6) In August 2022, the China Association for Science and Technology (CAST) initiated the establishment of the **Open Science Promotion Consortium (OSPC)**, including:

8 Academic Institutions: China National GeneBank(CNGB), Institute of Microbiology of the Chinese Academy of Sciences (CAS), Computer Network Information Center of CAS, among others.

10 Scientific Societies: American Chemical Society (ACS), Society of China University Journals, Royal Society of Chemistry (RSC), the International Association of Scientific, Technical and Medical Publishers (STM), among others.

7 Science Information Institutions: Institute of Scientific and Technical Information of China (ISTIC), National Science Library of CAS, Institute of Medical Information (IMI) & Library, Chinese Academy of Medical Sciences and Peking Union Medical College (CAMS & PUMC), among others.

14 International Publisher: World Scientific Publishing, Taylor & Francis Group, Springer Nature, among others.

7 Chinese Publishing Houses: Zhejiang University Press, Youke Publishing, Tsinghua University Press, among others.

8 Chinese Professional Information Service Institutions: China Academic Journals (CD Edition) Electronic Publishing House, Wanfang Data, China National Publications Import and Export (Group) Co., Ltd. (CNPIEC), among others.

Since its establishment, OSPC has organized a series of activities and conferences aimed at enhancing awareness and understanding of open science. For detailed information on specific conference names and dates, please refer to the official OSPC news section:<http://www.ospc.org.cn/news>

Meeting Name	Date
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Preparatory Meeting for Establishing the Open Science Promotion Consortium by the China Association for Science and Technology (CAST)	April 29, 2022
First Council Meeting of the Open Science Promotion Consortium	December 2, 2022
Second Council Meeting of the Open Science Promotion Consortium	June 18, 2023
Symposium on the Development of Open Access Ecosystem by the Open Science Promotion Consortium	January 19, 2024
Seminar on Green Open Access organized by the Open Science Promotion Consortium	January 25, 2024
Seminar on Enhancing the Exchange of Chinese Scientific Research Achievements and International Influence	March 12, 2024
2024 Annual Council Meeting of the Open Science Promotion Consortium	September 25, 2024

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES

If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

The actions undertaken include:

1. **Establishing Open Innovation Platforms:** The Ministry of Science and Technology (MOST) funds projects to establish open innovation platforms, promoting collaboration among industries, academia, and research institutions, as well as facilitating knowledge sharing. Additionally, MOST supports research on open science policies to explore effective implementation of open science principles. <https://pms.most.gov.cn/>
2. **Mandating Open Access for Funded Research:** The National Natural Science Foundation of China (NSFC) requires that research outcomes funded by the NSFC be published in open-access journals, ensuring free public access. Furthermore, the NSFC encourages researchers to share data during the research process to enhance transparency and reproducibility in scientific research. <https://www.nsfc.gov.cn/csc/20313/20331/20508/index.html>
3. **Developing Open Educational Resources (OER):** The Ministry of Education funds projects to develop open educational resources, making them freely available to teachers and students nationwide. It also supports academic exchange activities both domestically and internationally, promoting the dissemination and sharing of scientific knowledge.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES

If yes

please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

does it include any of the following open science elements:

- open access to scientific knowledge
- open science infrastructure
- open engagement of societal actors
- open dialogue with other knowledge systems

Please refer to the attached document for the relevant policy information.

The aforementioned policies and initiatives encompass various elements of open science, aiming to promote its implementation in China and to foster high-quality development in the nation's scientific and technological endeavors.

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

As of 2023, China has implemented a series of policies and measures to promote open science. These initiatives align with the definitions, values, and principles outlined in UNESCO's 2021 Recommendation on Open Science—such as transparency, inclusiveness, reproducibility, collaboration, and knowledge sharing. Specific examples of key elements of open science within China's national policies or policy frameworks are detailed in the attached table.

- 2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Yes, China has introduced specific policy tools to advance the development of open science. One significant initiative is the **International Open Access Journals Recommended List (OARL)**, accessible at <http://oarl.org.cn>. Launched by China National Publications Import & Export (Group) Corporation in collaboration with Tsinghua University Library and the Society of China University Journals, the OARL serves as a guiding directory of international open access journals. By referencing international evaluation standards and leveraging global information resources, the OARL aims to provide Chinese researchers with authoritative recommendations for high-quality open access journals. The directory adheres to the fundamental principles of responsibility, transparency, dynamism, and collaboration, employing a review mechanism that combines expert qualitative assessments with quantitative metrics. This ensures that the selected journals meet high standards in publishing norms, quality control, and open practices, thereby offering valuable guidance to Chinese researchers in identifying and choosing reputable open access journals.

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

YES

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

China has made significant progress in promoting the core elements of open science—such as open access, data sharing, collaboration, and inclusiveness—through a series of related policies, plans, and actions. Notable initiatives include:

Measures for Managing Scientific Data (2018): Issued by the State Council, these measures emphasize the importance of strengthening and standardizing scientific data management to ensure data security and promote open sharing.

Administrative Measures for the National Scientific and Technological Resource Sharing Service Platform: Released by the Ministry of Science and Technology of China, this policy aims to regulate the management of national platforms that facilitate the sharing of scientific and technological resources, thereby enhancing resource utilization efficiency and fostering innovation and entrepreneurship.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

YES

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#), included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

Here are five notable initiatives in China that promote open science and public participation:

1)Developer Alliance – CSDN Developer Community: A leading IT technology exchange platform in China, offering blogs, Q&A forums, training, and other services. <https://www.csdn.net/>.

2)China Birdwatching Record Center – Nationwide "Bird Monitoring Project": the center encourages public participation in bird monitoring, collecting birdwatching data to aid in avian conservation efforts. <https://t.yynet.cn/baijia/32801369.html>.

3)Chinese Academy of Sciences (CAS) – Public Science Day: An annual event where research institutions open their doors to the public, is to disseminate scientific knowledge and foster public understanding of science. https://www.cas.cn/cm/202105/t20210519_4788763.shtml?from=timeline.

4)Institute of Botany, CAS – Chinese Field Herbarium: Established by the Institute of Botany, CAS, this herbarium collects and displays plant specimens for public education and research purposes. <https://www.cfherb.ac.cn/>.

5)National Astronomical Observatories, CAS – Public Supernova Search Project (PSP): Initiated by the National Astronomical Observatories, CAS, this project invites public participation in the search and study of supernovae. <https://nadc.china-vo.org/pspl>.

2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: (ii) 17.i, (iii) 18.i, (iv) 19.c)

YES

If yes or under development, please provide details and links as relevant.

China has implemented several policies to promote open science, emphasizing open access, data sharing, collaboration, and inclusivity. Notable examples include:

1)Regulations on the National Natural Science Fund: Issued by the State Council, these regulations emphasize strengthening intellectual property management of funded research outcomes, supporting open access to research results, and promoting their application. https://www.gov.cn/gongbao/2024/issue_11726/202411/content_6989773.html.

2)Administrative Measures for the Open Sharing of National Major Scientific Research Infrastructure and Large Scientific Research Facilities: Jointly released by the Ministry of Science and Technology, the National Development and Reform Commission, and the Ministry of Finance, these measures state that, in principle, scientific research facilities and instruments should be openly shared with society. They are to provide services to universities, research institutes, enterprises, social R&D organizations, and individuals, with particular support for innovation, entrepreneurship, and the development of small and medium-sized enterprises.

2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES

If yes, please provide details and links as relevant.

The National Natural Science Foundation of China (NSFC) has issued a policy titled “National Natural Science Foundation of China Policy Statement on Open Access to

Research Publications from Its Funded Projects”.

<https://www.nsfc.gov.cn/csc/20313/20331/20508/index.html>.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

YES

If yes, please provide details and links as relevant.

In response to UNESCO's "Recommendation on Open Science" and to actively integrate China into global open science practices, the China Association for Science and Technology (CAST) initiated the establishment of the Open Science Promotion Consortium (OSPC). The aim of OSPC is to enhance the open sharing of scientific literature and data, thereby fostering openness, trust, and collaboration with the international scientific community. The official website of the OSPC is: <http://www.ospc.org.cn/>.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.i](#), [23.c](#))

YES

If yes or under development, please provide details and links as relevant.

CCOS and the National Science Library of the Chinese Academy of Sciences have developed a Open Science Readiness Digital Twin Dashboard. <http://os.las.ac.cn/zh-CN/research>

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers.

In 2024, China's total expenditure on Research and Experimental Development (R&D) surpassed ¥3.6 trillion, reaching ¥3.613 trillion, an increase of 8.3% compared to the previous year. The R&D expenditure intensity, defined as the ratio of R&D spending to Gross Domestic Product (GDP), rose to 2.68%, up by 0.10 percentage points from the previous year. Specifically, funding for basic research amounted to ¥249.7 billion, marking a 10.5% increase, and accounting for 6.91% of the total R&D expenditure, an improvement of 0.14 percentage points.

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

As of December 2024, China's internet users reached 1.108 billion, with an internet penetration rate of 78.6%. The number of online payment users was 1.029 billion, and online shopping users totaled 974 million. Both online retail sales and mobile payment penetration rates

continue to lead globally. These statistics were released on January 17, 2025, in the 55th Statistical Report on China's Internet Development by the China Internet Network Information Centre (CNNIC). https://www.cac.gov.cn/2025-02/18/c_1741582058470997.htm

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\)](#) [18.c](#))

YES

If **yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

- 1) **China Education and Research Network (CERNET):** The official website is <https://www.edu.cn/>.
- 2) **China Science and Technology Network (CSTNET):** The official website is <https://www.cstnet.cn/>.

- 3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\)](#) [18.d](#), [18.e](#), [18.k](#), [18.l](#), [18.m](#))

YES

If **yes or under development**, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
 - 1) **World Data Centre for Microorganisms (WDCM):** A non-commercial organization serves as a global platform for microbiological data management. <https://www.wdcm.org/>.
 - 2) **The National Tibetan Plateau Data Center (TPDC):** A non-commercial entity dedicates to collecting, managing, and sharing scientific data related to the Tibetan Plateau and surrounding regions. <https://data.tpdac.ac.cn/home>.
- community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
 - 1) **National Advanced Platform for Science and Technology Information and Communication:** <https://napstic.cn>.
 - 2) **National Science and Technology Report Service System:** <https://www.nstrs.cn/index..>
 - 3) **NSFC Basic Research Knowledge Base:** <http://or.nsfc.gov.cn/>.
 - 4) **CAS Institutional Repository Grid:** <http://www.irgrid.ac.cn/>.
 - 5) **National Social Science Academic Journal Database:** <https://www.nssd.cn/6>. <https://chinaxiv.org/home.htm>.
 - 6) **ChinaXiv: CAS Preprint Platform for Scientific Papers:** <https://chinaxiv.org/home.htm>.

Please note that all these facilities are non-commercial in nature.

- platforms for exchanges and co-creation of knowledge between scientists and society

- 1) **Beijing Electron Positron Collider (BEPCII):** <http://bepclab.ihep.cas.cn/>
- 2) **Jiangmen Underground Neutrino Observatory (JUNO):** <http://www.ihep.cas.cn/dkxzz/juno/>
- 3) **Heavy Ion Research Facility in Lanzhou (HIRFL):** https://www.gdlhz.ac.cn/dkxzz_02/
- 4) **Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST):** www.lamost.org
- 5) **Five-hundred-meter Aperture Spherical Radio Telescope (FAST):** <http://fast.bao.ac.cn>

Please note that all these facilities are non-commercial in nature.

- community-based monitoring and information systems to complement national, regional and global data and information systems.

- 1) **Global Open Taobao Data:** <https://tianchi.aliyun.com/dataset>
- 2) **Tencent Data Center:** <https://cloud.tencent.com/developer/column/2241>
- 3) **Huawei Cloud Data Center:** <https://activity.huaweicloud.com/qbitai.com+5developer.alibaba.com+5m.thepaper.cn+5>

Please note that all these platforms are non-commercial in nature.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\) 18.e](#), [18.f](#))

YES

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open scienceⁱ, stage of research, accessibility to all the relevant users in the region or internationally).

Name: World Data Centre for Microorganisms (WDCM)

Website: <https://wdcm.org/>

Establishment Date: Originally established in the 1960s, WDCM was officially relocated to the Institute of Microbiology, Chinese Academy of Sciences, in 2010.

Language: As an international data center, WDCM primarily uses English for communication and information dissemination, ensuring that users worldwide can access and utilize its resources without barriers.

Target Audience and Beneficiaries: WDCM serves a global community of microbiologists, data scientists, biotechnology experts, as well as related research institutions and industries. Researchers, students, policymakers, and the public with an interest in microbial resource data can conveniently access the necessary information through WDCM's platform, thereby promoting scientific research, technological innovation, and societal development.

Governance/Community Protocols: WDCM operates under the World Federation for Culture Collections (WFCC) of the International Union of Biological Sciences (IUBS) and the Microbial Resources Centers (MIRCEN) of UNESCO. It ensures data accuracy, completeness, and security by establishing and implementing a series of data management, sharing, and usage standards. Additionally, WDCM actively collaborates with microbial resource preservation institutions and researchers worldwide to promote open sharing and cooperative research of microbial resource data.

Scope: Microbiology, biotechnology, bioinformatics, and related fields.

Alignment with Open Science Elements: Open data sharing, collaborative research, knowledge production, and dissemination.

Research Stages Covered: WDCM's resources span the entire research spectrum, from basic research to applied development.

Accessibility: The WDCM platform is open and accessible to all relevant users, regardless of their region or country. Through this platform, users can easily browse, search, and download the microbial resource data they need.

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: [\(iii\) 18.g](#))

YES

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

In alignment with the principles and implementation plan of the UNESCO Recommendation on Open Science, the Alliance of National and International Science Organizations for the Belt and Road Regions (ANSO), the Chinese Academy of Sciences (CAS), UNESCO, and the African Academy of Sciences have jointly established the African "Open Science and Innovation Hub." This hub is based at the Sino-Africa Joint Research Center of the CAS. For the hub project, a budget of \$1 million has been allocated for 2025. Several Global South countries and ANSO members will share their achievements and experiences in the fields of open science and innovation. For more information, please visit: <https://www.anso.org.cn>.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

YES

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

1. **China Association of University Science and Technology Periodicals:** Led or participated in the development of various policies, standards, guidelines, and research projects, including collaborating with the International Association of Scientific, Technical and Medical Publishers (STM) to compile the "Open Access Publishing in China (2022)" report, and leading the formulation of the "Guidelines for Open Access in Scientific Journals." For more information, please visit: <http://m.cujs.org.cn/>
2. **National Science Library, Chinese Academy of Sciences:** Established the **GoOA** platform, which adheres to high-quality OA journal evaluation standards and OA paper integration usage norms. It aggregates high-quality OA journals and papers globally, further enhancing the value of open communication for OA papers and journals. For more information, please visit: <http://gooa.las.ac.cn/paper/#/home>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

- 1) **National Science and Technology Infrastructure Center:** Engages in national science and technology resource sharing services. For more information, please visit: <https://www.escience.org.cn/nsti/index>.
- 2) **National Data Administration:** Establishes data resource classification and hierarchical management systems, coordinates the confirmation and authorization of public data rights, and promotes interconnectivity of information resources across industries and departments. For more information, please visit: https://www.nda.gov.cn/sjj/index_pc.html.

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

YES

If yes, please provide more information and links as relevant.

- 1) **Fudan University:** Fudan University has been a pioneer in data science education and research. In 2007, it launched data science seminars, followed by the enrollment of doctoral students in data science in 2010. By 2013, the university introduced a graduate course titled "Data Science," and in 2014, it initiated the Data Scientist Training Camp. In 2015, Fudan officially began admitting graduate students in the data science program and offered a second bachelor's degree in the field. For more details, please visit: <https://sds.fudan.edu.cn/>.
- 2) **Tsinghua University:** Tsinghua University established the Tsinghua- Qingdao Data Science Research Institute in 2014 to meet the urgent needs of the industry and launched

a multidisciplinary big data master's program. For more information, please visit: <https://www.tsinghua.edu.cn/info/1182/38020.htm>.

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES

The National Science Library of the Chinese Academy of Sciences launches an Open Science Microvideo Series. For more information, please visit:

https://mp.weixin.qq.com/s?_biz=MzI3Mzg5MjY3NQ==&mid=2247490691&idx=1&sn=3777621fb20870def30b41868c4ca839&chksm=eb1d08afdc6a81b927caf474255b7a7fb705c776e90aa025a4f75a005ffc17373144e99db67e&scene=27

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

YES

If yes, please provide more information and links as relevant.

- 1) **The Institute of Neuroscience (ION) of the Chinese Academy of Sciences** has organized multiple "Brain and Intelligence Science" public outreach activities. For more information, please visit: <http://www.cebsit.cas.cn/cxwh/kphd/>.
- 2) **On December 25, 2024, China revised and adopted the "Law of the People's Republic of China on Science and Technology Popularization (2024 Revision)," which took effect upon promulgation.** This law promotes the dissemination and promotion of new technologies and knowledge across all sectors of society, encourages various innovation entities to engage in science popularization activities related to new technologies and knowledge, and supports the application of new technologies in these efforts. It aims to guide society in correctly understanding and utilizing scientific and technological achievements, thereby creating a favorable environment for their application. Additionally, the law mandates the establishment of an open and shared national repository of science popularization resources and public service platforms, facilitating the co-construction and sharing of such resources across society. It also enhances volunteer service systems and frameworks for science popularization, supports volunteers in conducting these services, and strengthens training and supervision mechanisms.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i](#))

YES

If yes, please provide more information and links as relevant.

The General Office of the State Council issued the "Guiding Opinions on Improving the Evaluation Mechanism for Scientific and Technological Achievements" (Guobanfa [2021] No. 26), emphasizing the development and improvement of a national database for scientific and technological achievements. This includes formulating promotion lists tailored to various application needs and ensuring that information on non-confidential scientific and technological achievements, supported by public funds, is disclosed in accordance with regulations. For more information, please visit:

https://www.gov.cn/gongbao/content/2021/content_5631817.htm.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

YES

If yes, please provide details and links as relevant.

- 1) **Law of the People's Republic of China on Scientific and Technological Progress:**

This law encourages the open sharing of research data and promotes the societal application of scientific research outcomes. For more information, please visit:

https://www.gov.cn/xinwen/2021-12/25/content_5664471.htm.

- 2) **Several Opinions on Further Strengthening the Construction of Scientific Research Integrity:** This document requires researchers to publicly disclose their research data and methodologies, enhancing transparency and reproducibility. For more information, please visit: https://www.gov.cn/zhengce/2018-05/30/content_5294886.htm.

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES

If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.

- 1) **Law of the People's Republic of China on Progress of Science and Technology:** This law explicitly encourages the open sharing of research data and promotes the societal application of scientific research outcomes. https://www.gov.cn/xinwen/2021-12/25/content_5664471.htm.
- 2) **Regulations on Post-Subsidy Management for National Science and Technology Programs and Special Funds:** Jointly issued by the Ministry of Finance and the Ministry of Science and Technology, these regulations encourage stakeholders to establish reward policies for open access to scientific knowledge.

- 5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20](#))

NO

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

YES, at the national level. YES

If **yes**, please indicate which of the recommended actions they address and provide more information and links as relevant:

In the realm of scientific research, China has implemented several initiatives to promote open science across various stages: In the **Research Infrastructure Sharing stage**, the government issued the "Measures for the Open Sharing of National Major Scientific Research Infrastructure and Large Scientific Instruments," aiming to enhance the accessibility and utilization of significant research facilities and instruments. **CAS Initiatives:** The Chinese Academy of Sciences (CAS) established the "CAS Major Scientific and Technological Infrastructure Sharing Service Platform," facilitating the open sharing of its major scientific facilities. In the **Open Access stage to Scientific Data**, National bodies, along with institutions like CAS and the Chinese Academy of Agricultural Sciences, have formulated data management measures to promote data openness and utilization. In the **Data Accessibility stage**, centers such as the **Five-hundred-meter Aperture Spherical Radio Telescope (FAST)** have made their data publicly available to the global scientific community. In the **Dissemination stage of Research Outcomes**, organizations like CAS and the National Natural Science Foundation of China (NSFC) have issued statements advocating for open access, requiring that research papers be openly stored and accessible. In the **repositorying stage of research outputs**, CAS, in collaboration with universities, has developed institutional repositories (IRs), and the NSFC continues to build the Open Access Repository (OAR) platform to facilitate open access to research findings. In the stage of **Open Education**, higher education institutions are leveraging Massive Open Online Courses (MOOCs) and other platforms to provide open educational resources, enhancing the accessibility of knowledge. In the **International Scientific Cooperation**, initiatives like the Digital Earth (DDE) and the Big Earth Data Center for Sustainable Development Goals (SDG) are engaging in global collaborations, serving as exemplary cases of open science practices.

Collectively, these efforts underscore China's commitment to fostering open science through infrastructure sharing, data openness, accessible research outcomes, open education, and international cooperation.

- promotion of pre-prints

In China, several institutions have established preprint platforms to facilitate the rapid dissemination of scientific research:

- 1) **ChinaXiv:** Developed by the Chinese Academy of Sciences (CAS), this platform serves as a repository for preprints and open-access papers across various scientific disciplines. For more information, please visit: <https://chinaxiv.org/home.htm>.

- 2) **National Preprint Platform:** Established by the Institute of Scientific and Technical Information of China, this platform aims to provide a centralized repository for preprints across multiple scientific fields. For more information, please visit: <https://sinoxiv.napstic.cn/>.
- 3) **biomedRxiv:** Managed by the Chinese Academy of Medical Sciences, biomedRxiv is a preprint system dedicated to biomedical sciences, supporting the free submission and publication of biomedical research papers. For more information, please visit: <https://www.biomedrxiv.org.cn/index>.

These platforms aim to promote open access and timely sharing of research findings within the scientific community.

- exploration of open peer-review practices

China Scientific Data (CSD) is a bilingual (Chinese and English) online journal dedicated to the open sharing and citation of scientific data. It aims to promote the long-term preservation of scientific data, enhance data asset management, explore effective evaluation mechanisms for data-related work, advance the development of data science, and ensure that scientific data are findable, accessible, interoperable, and reusable (FAIR principles). For more information, please visit their website: <http://www.csdata.org/p/static/33/>.

- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects

- 1) **China Science Communication** (科普中国, www.kepuchina.cn), a national science popularization platform dedicated to disseminating scientific knowledge, enhancing public science literacy, and promoting innovation-driven development. For more information, please visit: <https://www.kepuchina.cn/>

- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
 - 1) The China Association for Science and Technology (CAST), the Ministry of Science and Technology (MOST), and 11 other government departments jointly issued the **"Measures to Support Women in Science and Technology to Play a Greater Role in Innovation"** (Guokefacai [2021] No. 172). These measures aim to further stimulate the innovative potential of female scientists and enhance their contributions to innovation-driven development.
 - 2) In October 2015, CAST launched the **"Young Talent Support Program,"** which selectively provides support through its affiliated national academic societies or academic alliances to foster young scientific talent.
- promoting open innovation practices.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

- 1) In line with the UNESCO Recommendation on Open Science, in order to promote global cooperation on ST&I, and work towards an open, fair, just and non-discriminatory environment for global science and technology development, and advance the sound development of global open science, the African Union, Brazil, China and South Africa put forward the Initiative on International Cooperation in Open Science in 2024 G20 Leaders' Summit.
- 2) In alignment with the principles and implementation plans outlined in **UNESCO's Recommendation on Open Science**, the **Alliance of National and International Science Organizations for the Belt and Road Regions (ANSO)**—an international scientific organization based in Beijing, with the Chinese Academy of Sciences (CAS) as a permanent member of its governing council—collaborated with CAS, UNESCO, and the African Academy of Sciences to establish the African "Open Science and Innovation Hub." This initiative is anchored in the China-Africa Joint Research Center, affiliated with CAS. For the Hub project, a budget allocation of \$1 million annually has been earmarked for 2025. Several Global South countries and ANSO members will share their achievements and experiences in the realms of open science and innovation. Additionally, ANSO has budgeted \$1.2 million for 2025 to support the joint research project **"Open Science for Sustainable Development."** The primary audience for this initiative comprises research institutions and scientists from countries along the "Belt and Road" route. Each project will be led by an ANSO member, collaborating with counterparts from other member countries to conduct interdisciplinary research. Furthermore, three "Open Science Training Courses" will be organized annually. For more information, please visit: <http://www.anso.org.cn/>.
- 3) To actively integrate China into global open science practices, enhance the open sharing of scientific literature and data, and foster openness, trust, and cooperation with the international scientific community, the China Association for Science and Technology (CAST) initiated the formation of the **Open Science Promotion Consortium (OSPC)**. Currently, OSPC comprises numerous domestic and international research institutions, scientific organizations, publishing entities, information service providers, and other member units. Notably, several renowned overseas publishing houses, the International Association of Scientific, Technical, and Medical Publishers (STM), and prominent international academic groups are among its members. For more information, please visit: <http://www.ospc.org.cn/>.

These initiatives underscore China's commitment to advancing open science and fostering international collaboration in scientific research and innovation.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b](#), [22.e](#), [22.f](#))

YES

If yes, please provide links as relevant and more information.

Since 2011, the Chinese Academy of Sciences (CAS) has organized the annual **China Open Access Week** to promote knowledge sharing and utilization, fostering the development of scientific research towards greater openness, equity, transparency, and sustainability.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=21>)

YES

If yes, please provide links as relevant and more information.

In 2014, China played a pivotal role in establishing the **Consortium for Reliability and Reproducibility (CoRR)**, an international initiative aimed at enhancing the reliability and reproducibility of neuroimaging research. For more information, please visit:

https://fcon_1000.projects.nitrc.org/indi/CoRR/html/index.html

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO

i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).

ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

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- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
- xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.